

UNIVERSITY OF SOUTHAMPTON

DEPARTMENT OF ECONOMICS

SCHOOL OF ECONOMIC, SOCIAL & POLITICAL SCIENCES

DISSERTATION: RESEARCH TOPICS

FINAL REPORT

Topic 1: Returns to Education

34840559

Presented for B.Sc. in Economics and Actuarial Science

23rd April 2026

I declare that this report is my own work and that all material from published or unpublished sources has been properly acknowledged. I also declare that I have used AI in accordance with the University of Southampton guidelines to assist with the initial literature search and to improve clarity and grammar. All econometric analysis, interpretation, and conclusions are my own work.

Signed: 34840559 23rd April 2026

Table of Contents

1. Introduction	1
2. Economic Analysis.....	2
2.1. Dataset.....	2
2.2. Econometric Models.....	6
3. Empirical Results	8
3.1. Ordinary Least Squares (OLS) Results.....	8
3.2. Instrumental Variable (IV) Results and Comparison with the Literature....	11
3.3. Data Issues and Robustness	14
3.4. IV Limitations	16
4. Conclusion	18
5. Appendix	19
5.1. Summary Statistics	19
5.2. Wage Distribution Summary and Top-Coding Assessment.....	20
5.3. Figures.....	20
5.4. OLS Estimation Code.....	20
5.5. First-Stage and IV (2SLS) Estimation Code.....	21
5.6. Instrument Strength Across Specifications	21
6. References	22

1. Introduction

Education plays a central role in labor markets. Many studies show that well-educated individuals earn higher wages and experience better labour market outcomes. Schultz (1961) views education as an investment in 'human capital', referring to the skills and knowledge through education that increases labour productivity and future earnings. However, interpreting this positive association between education and income as a causal relationship remains challenging. Griliches (1977) demonstrates that estimated returns to schooling can be biased by omitted-variable bias, in which unobserved factors, such as innate ability and family background, affect both education and earnings.

This study aims to estimate the causal return to education by examining the relationship between years of schooling and earnings. In particular, it addresses endogeneity in education by comparing ordinary least squares (OLS) and instrumental-variable (IV) estimates, using proximity to college as an instrument. The results show that education is positively associated with earnings, with OLS estimates suggesting a return of approximately 6 percent per additional year of schooling. However, the IV estimates are of similar magnitude but not statistically significant, likely reflecting the relatively weak instrument. These findings suggest that, while education is an important determinant of earnings, identifying its causal influence is complex.

2. Economic Analysis

2.1. Dataset

This study uses data from the National Longitudinal Survey of Young Men (NLSYM) in 1976 to examine the relationship between education and earnings. The dataset contains information on wages, years of schooling, labour market experience, and demographic characteristics for a sample of 3,010 men aged 24 – 34. Although NLSYM76 is a longitudinal survey, this analysis uses only the 1976 wave and treats the data as cross-sectional.

The dataset is well suited to analysing the returns to education, as it contains the key variables required to implement the Mincer (1974) earnings framework, including wages (and its quadratic), years of schooling, and labour market experience. In addition, it includes demographic and background characteristics that allow for the consideration of potential sources of bias in the estimation. Finally, the sample also exhibits variation in experience, demographic characteristics, and background variables, providing sufficient heterogeneity for empirical analysis.

The dependent variable in this analysis is the logarithm of hourly wages ($\ln wage$), where the log transformation is applied to address the positive skewness of wages, as shown in Figure 1 and 2. The key explanatory variable is years of schooling ($educ$), which measures the quantity of education attained. As shown in Figure 3, education is concentrated around key levels such as 12 and 16 years, reflecting common schooling thresholds.



Figure 1: Distribution of Hourly Wages

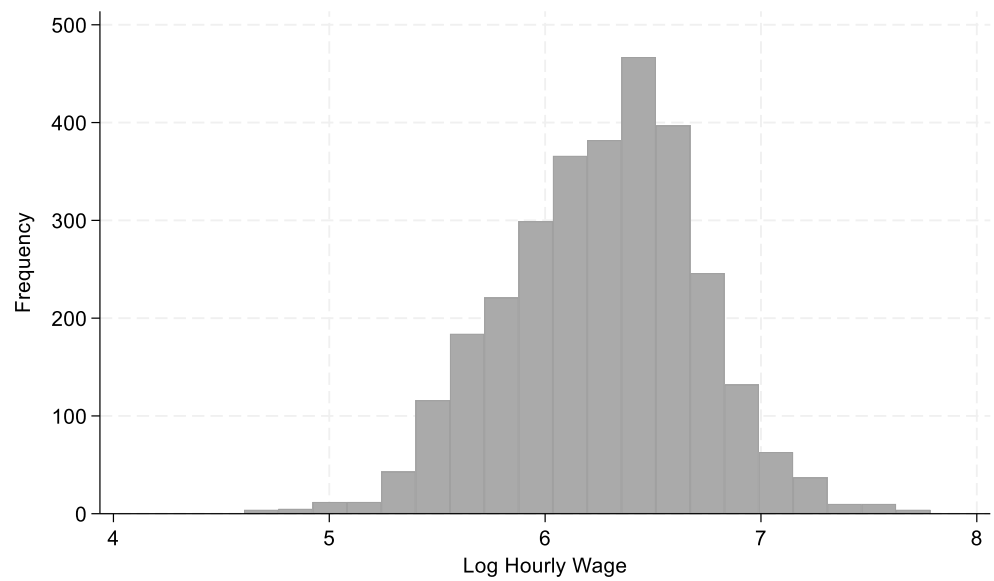


Figure 2: Distribution of Log Hourly Wages

Labour market experience (*exper*) and its square (*expersq*) are included to capture the concave relationship between experience and wages predicted by the Mincer (1974) model. As shown in Table 1, individuals have on average

approximately 8.9 years of labour market experience, with a standard deviation of 4.1, indicating substantial variation across the sample. Experience is measured as potential labour market experience, calculated as age minus six years of schooling. Additional control variables are included to account for observable differences across individuals. These include demographic and geographic characteristics, namely race (black), marital status (married), residence in an urban area (smsa) and in the Southern region of the United States (south). Moreover, measures of ability, such as intelligence quotient (IQ) and Knowledge of the World of Work (KWW), are also contained. Table 1 shows an average IQ of 102 and KWW of 33.5, indicating meaningful variation in ability across individuals. Finally, family background is captured through parental education, reflecting socioeconomic conditions that may influence both educational attainment and earnings.

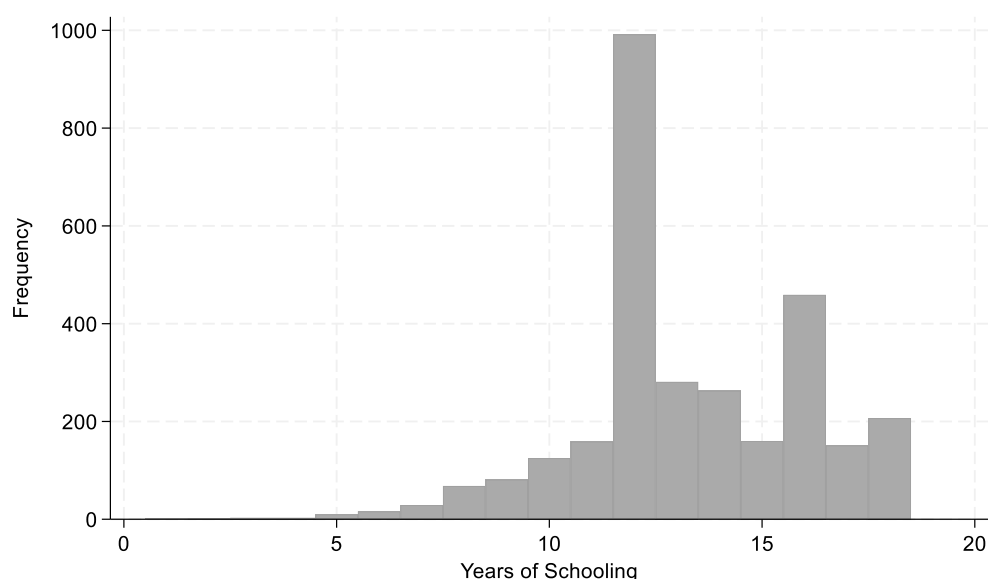


Figure 3: Distribution of Years of Schooling

Despite its richness, the dataset has several limitations that may affect the reliability of the estimates. These issues are discussed further in the Section 3.3 (Data Issues and Robustness).

Variables	Observations	Mean	Std. Dev.	Min	Max
educ	3,010.00	13.26	2.68	1.00	18.00
fatheduc	2,320.00	10.00	3.72	0.00	18.00
Black	3,010.00	0.23	0.42	0.00	1.00
Smsa	3,010.00	0.71	0.45	0.00	1.00
South	3,010.00	0.40	0.49	0.00	1.00
Wage	3,010.00	577.28	262.96	100.00	2,404.00
KWW	2,963.00	33.54	8.61	4.00	56.00
IQ	2,061.00	102.45	15.42	50.00	149.00
exper	3,010.00	8.86	4.14	0.00	23.00
lwage	3,010.00	6.26	0.44	4.61	7.78
expersq	3,010.00	95.58	84.62	0.00	529.00
married_	3,003.00	0.71	0.45	0.00	1.00

Table 1. Summary Statistics

Notes: wage is hourly wage (USD); lwage is the natural logarithm of wage; educ is years of schooling; exper is potential labour market experience; expersq is experience squared; black, smsa, south, and married are binary indicators; IQ and KWW proxy ability; motheduc and fatheduc measure parental education.

2.2. Econometric Models

The empirical analysis is based on the standard Mincer (1974) wage equation, which relates the logarithm of wages to years of schooling, labour market experience, and its square, and is given by

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{exper}_i^2 + u_i$$

where β_1 represents the return to an additional year of schooling. To account for observable differences across individuals, the model is extended to include additional control variables

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{exper}_i^2 + X_i' \gamma + u_i$$

where X_i includes demographic, geographic, ability, and family background variables. OLS is used to estimate both in the Mincer baseline form and the extended model with additional control, providing a benchmark for estimating the return to education.

A key concern is that education may be endogenous, as it can be correlated with unobserved factors such as ability and motivation. Although proxy variables such as IQ and KWW are included, they may not fully capture these effects. To address this issue, the analysis applies an instrumental variable (IV) approach using two-stage least squares (2SLS). In the first stage, education is regressed on the instrument and all exogenous variables

$$\text{educ}_i = \pi_0 + \pi_1 Z_i + X_i' \delta + v_i$$

where Z_i denotes the instrumental variable, X_i includes all exogenous control variables in the model, and δ denotes coefficients on controls in the first stage. In

the second stage, the predicted values of education, $\widehat{educ}_i$, are used in the wage equation

$$\log(wage_i) = \beta_0 + \beta_1 \widehat{educ}_i + X_i' \gamma + u_i$$

where γ denotes coefficients on controls in the second stage.

Proximity to college (nearc4) is used as an instrument for education, capturing variation in schooling driven by differences in access to higher education. This indicator tells whether an individual grew up near a four-year college and captures exogenous variability in education due to geographical differences in access to higher education, which affects the cost of university education (Card, 1995). For this tool to be valid, it must satisfy the exogeneity condition, meaning that geographic proximity to college must not correlate with unobservable factors such as ability or family influence, and primarily influences schooling decisions through differences in access and cost, rather than directly affecting labour market outcomes (Card, 1995).

3. Empirical Results

3.1. Ordinary Least Squares (OLS) Results

Table 2 presents the OLS estimates of the Mincer wage equation across three different model specifications. In column (1), the baseline model includes the years of schooling, work experience, and its square. The estimated return from education is 0.093, which is statistically significant and implies that each additional year of schooling is associated with a wage increase of 9.3%.

Experience has a positive coefficient, while its square is negative, confirming a concave relationship in which wages rise with experience at a decreasing rate. The model explains 19.6% of the variation in wages.

In column (2), demographic and geographic control variables are added. The estimated return to education decreased to 0.072, suggesting that part of the initial estimate is driven by omitted variable bias related to observable characteristics. Individuals living in urban areas (smsa) earn approximately 17.2% higher wages, while those living in the South earn approximately 12.7% lower wages. Marital status is associated with approximately 14.3% higher wages, while being Black is associated with approximately 17.0% lower wages. These effects are economically significant and including these control variables increase the model's explanatory power, with R^2 increasing to 0.311.

In column (3), the model incorporates variables representing ability and family circumstances. The estimated benefit from education decreased to 0.060, suggesting that factors related to ability further explain the relationship between

education and wages. Both IQ and KWW have positive and statistically significant coefficients, suggesting that cognitive ability and job-related knowledge are important determinants of wages. For example, a one-point increase in IQ is associated with a 0.2% increase in wages, while a similar increase in KWW corresponds to a rise of 0.4%, highlighting the importance of ability. Parental education shows contrasting effects: maternal education is positive and statistically significant, with each additional year associated with a 0.9% increase in wages, while paternal education is not statistically significant. This suggests that maternal education may capture aspects of family background that directly influence earnings, whereas the effect of paternal education is largely absorbed by other variables such as individual education and ability.

However, the inclusion of these variables leads to a significant reduction in sample size, from over 3,000 to just 1,603 observations due to missing data, raising concerns about potential sampling bias. The decline in R^2 to 0.237 partly reflects this reduced sample rather than weaker explanatory power. Moreover, IQ and KWW are imperfect proxies for ability and may be measured with error.

Overall, the decline in the estimated return to education from 9.3% to 6.0% provides strong evidence of omitted variable bias in simpler specifications. While the extended model improves the analysis, OLS estimates may still overstate the true causal effect of education on wages.

Variables	Baseline (1)	Controls (2)	Full Model (3)
educ	0.093 (0.004)	0.072 (0.003)	0.060 (0.006)
exper	0.090 (0.007)	0.074 (0.007)	0.079 (0.011)
expersq	- 0.002 (0.000)	- 0.002 (0.000)	- 0.002 (0.001)
smsa		0.172 (0.015)	0.158 (0.022)
south		- 0.127 (0.015)	- 0.083 (0.020)
married_		0.143 (0.015)	0.142 (0.022)
black		- 0.170 (0.018)	- 0.078 (0.035)
IQ			0.002 (0.001)
KWW			0.004 (0.002)
motheduc			0.009 (0.004)
fatheduc			- 0.004 (0.004)
Constant	4.469 (0.069)	4.710 (0.067)	4.455 (0.119)
Observations	3,010	3,003	1,603
R-squared	0.196	0.311	0.237

Table 2. OLS Estimates of the Returns to Education (Standard errors in parentheses)

3.2. Instrumental Variable (IV) Results and Comparison with the Literature

Table 3 presents the first-stage regression results, where education is regressed on the instrumental variable and other controls. The first-stage results show that proximity to college is positively correlated with educational attainment. The estimated coefficient of `nearc4` is 0.230 and is statistically significant, implying that individuals who grew up near a four-year college have higher-than-average educational attainment. However, the corresponding F-statistic is 7.06, which is below the conventional threshold of 10, suggesting that the instrument may be relatively weak. The strength of an instrument is often assessed using the first-stage F-statistic, with low values indicating potential problems of weak identification (Bound, Jaeger, and Baker, 1995). A weak instrument may limit the reliability and precision of the IV estimates in the second stage.

Table 4 reports the second-stage IV estimates of the return to education. The coefficient on education is 0.061, suggesting that an additional year of schooling increases wages by approximately 6.1%. This result is very similar to the OLS estimate obtained in the fully specified model; however, it is not statistically significant and comes with a significantly larger standard error. This lack of precision is likely driven by the limited strength of the instrument, as a weak first-stage relationship reduces the variation in education used for identification and leads to low statistical power in the IV estimation. As a result, although the IV and OLS estimates are economically similar, the IV estimate may not be precisely estimated in this model.

Variables	First-Stage	Variables	First-Stage
nearc4	0.230 (0.086)	IQ	0.022 (0.003)
exper	- 0.568 (0.043)	KWW	0.112 (0.006)
expersq	0.013 (0.002)	motheduc	0.051 (0.016)
smsa	0.017 (0.091)	fatheduc	0.053 (0.014)
south	0.124 (0.083)	Constant	9.800 (0.413)
married_	0.210 (0.086)	black	0.713 (0.139)
Observations	1,603	R-squared	0.577

Table 3. First-Stage Regression of Education on Instrument (Standard errors in parentheses)

In contrast to most of the existing literature, where IV estimates are often larger than those from OLS, Card (1995) used geographic variation in proximity to college as an instrument for education, capturing differences in the cost of schooling, and found IV estimates of the return to education to be substantially higher than OLS estimates, typically by 25–60%. Similarly, Oreopoulos (2006) exploited changes in compulsory schooling laws in the UK and reported IV

estimates significantly above OLS, with returns increasing from around 8% to 14–17%. The author argued that this difference reflects the use of an instrument that identifies the impact of education on individuals whose schooling decisions are influenced by policy changes.

However, not all studies find that IV estimates differ substantially from OLS estimates. Angrist and Krueger (1991) used quarter of birth as an instrument for education, exploiting variation created by compulsory schooling laws in the United States. Individuals born earlier in the year tended to start school later and might leave school earlier upon reaching the minimum school-leaving age. Their IV estimates of the return to education were very similar to their corresponding OLS estimates and showed no statistically significant differences. The authors interpreted this finding as evidence that OLS estimates may not be severely biased by unobserved factors. Thus, while this instrument identifies differences in schooling among individuals affected by compulsory education laws, the estimated returns appear similar to the average return, resulting in only small differences between IV and OLS estimates. Overall, these findings suggest that the relationship between IV and OLS estimates depends critically on the choice and strength of the instrument, as well as the population whose educational decisions are affected.

Variables	IV (2SLS)	Variables	IV (2SLS)
educ	0.061 (0.093)	married_	0.142 (0.029)
exper	0.079 (0.054)	black	- 0.078 (0.076)
expersq	- 0.002 (0.001)	IQ	0.002 (0.002)
smsa	0.158 (0.023)	KWW	0.004 (0.011)
south	- 0.083 (0.022)	motheduc	0.009 (0.006)
fatheduc	- 0.004 (0.006)	Constant	4.454 (0.928)
Observations	1,603	R-squared	0.237

Table 4. IV Estimates of the Return to Education (Standard errors in parentheses)

3.3. Data Issues and Robustness

This section discusses potential data-related issues and assesses the robustness of the empirical results. In particular, several limitations may affect the reliability of the estimates, including missing data, measurement errors, and sample selection. To assess the robustness of the findings, the analysis considers multiple

model specifications, including the use of additional control variables and instrumental variables.

The first concern is the risk of measurement error in the educational attainment data. While years of schooling is generally considered a continuous variable, its distribution shows a clear concentration at key thresholds, particularly at 12 and 16 years of schooling, corresponding to high school and college completion. This pattern highlights the discrete and degree-based nature of the education system, suggesting that schooling may not fully reflect differences in basic skills or human capital. While this approach is standard, the results should be interpreted with caution.

Issues related to wage measurement and distribution are also relevant.

Measurement errors in wages may lead to attenuation bias and reduce the precision of the analysis. The wage distribution exhibits a positive skew, indicating the presence of significantly higher earners than the majority. The analysis applies a logarithmic transformation to wage values, resulting in a more symmetrical distribution and minimising the influence of extreme values.

The potential impact of top-coding is limited in this sample. As shown in Table 6 in the Appendix, although the maximum observed wage (2,404) exceeds the 99th percentile (1,442), only one observation reaches this level. This suggests that top-coding does not significantly distort the distribution or bias the results.

Work experience is not directly observed and is instead approximated using age and years of schooling. This measure does not account for unemployment or career interruptions, which may introduce error. Similarly, variables capturing ability and family background, such as IQ, test scores, and parental education, may be subject to measurement error and may contain missing observations. As shown in Table 2, the sample size declines substantially from 3,010 observations in the baseline model to 1,603 in the full specification, potentially introducing selection bias and affecting the reliability of the analysis.

Finally, these variables may not fully capture the constructs they are intended to measure. Ability cannot be fully represented by indicators such as IQ or knowledge test scores, while parental education may only partially reflect family circumstances and socioeconomic status. Additionally, geographic variables, such as urban residence and regional indicators, included to explain differences in labour market conditions, are relatively rudimentary and may not fully capture local differences in employment opportunities, wages, or the cost of living. Therefore, they can only partially control for income disparities between regions.

3.4. IV Limitations

Most importantly, the instrumental variable used in this study appears to be relatively weak. The first-stage F-statistic falls below the conventional threshold in the full model and only exceeds it when some control variables are removed, as shown in Table 5, indicating that the explanatory power of the instrument is sensitive to model specification. This suggests that proximity to college does not

provide a strong and stable source of independent variation in educational attainment. From an economic perspective, this result is intuitive, as educational decisions are influenced by many factors beyond geographic distance, including ability, family background, and individual preferences. Moreover, improvements in transportation and access to information have reduced the importance of proximity as a constraint on higher education decisions.

Model	Coefficient	F-stat	Observations
Baseline	0.600 (0.079)	58.02	3,010
Controls	0.327 (0.082)	15.70	3,003
Full Model	0.230 (0.086)	7.06	1,603

Table 5. Table X: Instrument Strength Across Specifications

Notes: Standard errors in parentheses. The F-statistic tests the significance of the instrument in the first stage.

In addition, the validity of the instrument relies on the assumption that proximity to college affects earnings only through education (Card, 1995). This assumption may be questioned if proximity is correlated with local economic conditions or family characteristics that directly influence wages. For example, families who prioritise education may choose to live near universities or in areas with higher-quality schooling, which can lead to better labour market outcomes for their children independently of education. These channels suggest that proximity to

college may be correlated with unobserved determinants of wages, thereby violating the exogeneity assumption.

Finally, the IV estimates should be interpreted within the Local Average Treatment Effect (LATE) framework, as they capture the return to education only for individuals whose schooling decisions are influenced by proximity to college, rather than for the entire population (Imbens and Angrist, 1994).

4. Conclusion

This study examines the return to education by estimating the relationship between years of schooling and income using OLS and IV approaches. Both methods show a positive relationship between education and wages. The OLS estimate indicates a return of about 6% per additional year of schooling after controlling for observable characteristics. The IV estimate is similar in magnitude at 6.1% but is not statistically significant due to large standard errors.

Several limitations are identified in this study. Data issues, such as measurement error and missing observations, may introduce bias and reduce accuracy. Years of schooling may not capture all relevant differences, while ability and family background are only imperfectly measured. Geographic and demographic controls may not fully reflect differences in labour market conditions, leaving some variation unobserved.

Secondly, the IV approach is constrained by a relatively weak instrument, which reduces its ability to precisely estimate the causal effect of education on income. In addition, the validity of the instrument may be questioned if proximity to college is correlated with unobservable factors that influence income. It is also important to interpret IV estimates within the Local Average Treatment Effect (LATE) framework, meaning that the results apply only to individuals whose educational choices are affected by the instrument.

Future research could incorporate richer measures of ability and non-cognitive skills, such as personality traits or motivation, which are not fully captured by proxies like IQ and KWW. Additionally, extending the analysis to different cohorts or more recent datasets would improve the generalisability of the results, as labour market conditions and the returns to education may vary over time and across groups.

5. Appendix

This appendix provides the STATA code used to generate the results in the report.

All regressions are estimated using robust standard errors.

5.1. Summary Statistics

```
summarize wage lwage educ exper expersq IQ KWW motheduc fatheduc  
married_ black smsa south
```

5.2. Wage Distribution Summary and Top-Coding Assessment

summarize wage, detail

	Observations	Mean	Std. Dev.	Min	Max	p99
wage	3,010	577.3	263.0	100	2,404	1,442

Table 6. Wage Distribution Summary

Notes: The table reports summary statistics for hourly wages (USD).

5.3. Figures

* Figure 1: Distribution of Hourly Wages

```
histogram wage, bin(20) frequency color(gs10) xtitle("Hourly Wage (USD)")  
ytitle("Frequency") title("Distribution of Hourly Wages")  
graphregion(color(white))
```

* Figure 2: Distribution of Log Wages

```
histogram l wage, bin(20) frequency color(gs10) xtitle("Log Hourly Wage")  
ytitle("Frequency") title("Distribution of Log Wages") graphregion(color(white))
```

* Figure 3: Distribution of Years of Schooling

```
histogram educ, discrete frequency color(gs10) xtitle("Years of Schooling")  
ytitle("Frequency") title("Distribution of Education") graphregion(color(white))
```

5.4. OLS Estimation Code

* Baseline model

```
reg l wage educ exper expersq, robust
```

* With controls

```
reg lwage educ exper expersq smsa south married_ black, robust
```

* Full model

```
reg lwage educ exper expersq smsa south married_ black IQ KWW motheduc  
fatheduc, robust
```

5.5. First-Stage and IV (2SLS) Estimation Code

* First-stage regression

```
reg educ nearc4 exper expersq smsa south married_ black IQ KWW motheduc  
fatheduc, robust
```

* IV estimation

```
ivregress 2sls lwage (educ = nearc4) exper expersq smsa south married_ black IQ  
KWW motheduc fatheduc, robust
```

5.6. Instrument Strength Across Specifications

* Baseline specification

```
reg educ nearc4 exper expersq, robust
```

```
test nearc4
```

* With controls

```
reg educ nearc4 exper expersq smsa south married_ black, robust
```

```
test nearc4
```

* Full specification

```
reg educ nearc4 exper expersq smsa south married_ black IQ KWW motheduc  
fatheduc, robust
```

test nearc4

6. References

Schultz, T.W. (1961). Investment in Human Capital. *The American Economic Review*, 51(1), pp.1–17. Available at: <http://www.jstor.org/stable/1818907>.

Griliches, Z. (1977). Estimating the Returns to Schooling: Some Econometric Problems. *Econometrica*, 45(1), pp.1–22. Available at: <https://doi.org/10.2307/1913285>.

Mincer, J.A. (1974). The human capital earnings function. In: J.A. Mincer, *Schooling, experience, and earnings*. New York: National Bureau of Economic Research, pp.83–96.

Card, D. (1995). Using geographic variation in college proximity to estimate the return to schooling. In: Christofides, L.N., Grant, E.K. and Swidinsky, R. (eds.) *Aspects of Labour Market Behaviour: Essays in Honour of John Vanderkamp*. Toronto: University of Toronto Press, pp.201–222.

Oreopoulos, P. (2006). Estimating Average and Local Average Treatment Effects of Education When Compulsory Schooling Laws Really Matter. *The American Economic Review*, [online] 96(1), pp.152–175. Available at: <https://doi.org/10.2307/30034359>.

Bound, J., Jaeger, D.A. and Baker, R.M. (1995). Problems with Instrumental Variables Estimation When the Correlation Between the Instruments and the Endogenous Explanatory Variable is Weak. *Journal of the American Statistical Association*, 90(430), p.443–450. Available at: <https://doi.org/10.2307/2291055>.

Angrist, J.D. and Krueger, A.B. (1991). Does Compulsory School Attendance Affect Schooling and Earnings? *The Quarterly Journal of Economics*, 106(4), pp.979–1014. Available at: <https://doi.org/10.2307/2937954>.

Imbens, G.W. and Angrist, J.D. (1994). Identification and Estimation of Local Average Treatment Effects. *Econometrica*, 62(2), pp.467–475. Available at: <https://doi.org/10.2307/2951620>.